

Basic (single-agent support bot design) — Hints

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Only 2 hints — work through them in order against your own written design before reading ahead. Each hint has 3 depth levels: **Basic** (the plain framing — what to ask, what the obvious first design looks like), **Intermediate** (the real architectural vocabulary and trade-offs), **Advanced** (the senior-level judgment call — what a naive design gets wrong, and what a stakeholder would push back on). Read Basic first even if you already know the vocabulary — it's the fastest way to see exactly what each deeper level adds.

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Hint 1 — What to ask, and the obvious first design

Basic Version

Before drawing anything, figure out three things: what happens when the bot gets it wrong (does a wrong answer cost money, embarrassment, or nothing much)? How many people will use it, and how often? And what does "unsure" actually mean — no docs found, a low-confidence guess, or a question that isn't even about the product?

The obvious first-draft design: one agent that can search the product docs and answer, with a rule like "if it can't find a good answer, hand off to a human."

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Intermediate Version

This is a retrieval-augmented generation (RAG) design — Doc08 material — wrapped in a single tool-calling agent (Doc11's simplest pattern: one agent, one or more tools, no multi-agent routing needed for a task this size). The two real pieces:

- A `search_docs` tool that retrieves over the product documentation (embeddings + vector search, or plain keyword search at very small scale — matching the tool to the actual traffic, not defaulting to the fanciest option available).
- A routing decision after retrieval: is there a real basis for an answer, or not? Three real ways to define "unsure," each with a different reliability:
- **Retrieval-score threshold** — if nothing scores above X similarity, treat it as unsure. Cheap and mechanical, but doesn't understand meaning.
- **LLM self-reported confidence** ("rate your confidence 1-10") — cheap to add, but models aren't reliably calibrated about their own confidence, so trusting this alone is risky.
- **A grounding check** — did the retrieved docs actually cover the specific thing asked, or is the model filling the gap with its own general knowledge? This is the one that actually protects against confidently stating something product-specific that isn't true.

State here is simple: per-conversation history plus the most recent retrieval result, so the agent's answer can be traced back to what it actually found.

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Advanced Version

The naive version of this design uses the LLM's own stated confidence to decide when to hand off — and that's exactly the piece a senior reviewer will push on. A model can sound completely confident while stating something the docs never said, because "sounding confident" and "grounded in retrieved text" are not the same signal. The safer check asks a narrower question: does at least one retrieved chunk actually support this specific claim — and treats an answer that leans on the model's general knowledge instead of the retrieved text as automatically unsure, no matter how fluent it sounds.

A stakeholder will also ask "why hand off to a human at all instead of just saying 'I don't know'" — the honest answer is that a real support flow needs an actual resolution path, not a dead end. So the design has to name where handoff sends the conversation (a queue? a live agent? a ticket?) and what travels with it — the user's question and what the bot already tried, so the human doesn't start from zero.

Cost/latency: RAG adds one retrieval step before generation. For a single-agent bot at modest traffic that's fine as-is. Caching repeated queries only becomes a justified addition once volume actually requires it — adding it now, for traffic nobody mentioned, is the exact "inspiration instead of matching" mistake Core Concepts warns about.

Difference between Basic, Intermediate, and Advanced: Basic names the questions to ask and sketches one clean agent-plus-retrieval design. Intermediate supplies the real vocabulary and three concrete ways to implement "unsure," each with a real trade-off. Advanced is the judgment call an interviewer is actually listening for: which of those three "unsure" definitions is trustworthy versus which one just sounds trustworthy, and what the handoff needs to carry to actually be useful downstream.

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Hint 2 — The plan, and almost the whole design

Basic Version

figure out:

- what happens if the bot answers wrong?
- how many users, how often?
- what exactly counts as "unsure"?

design:

- one agent
- one tool: search the product docs
- rule: if no good match found in the docs, hand off to a human

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Intermediate Version

Agent: single support agent (Doc11's simplest pattern – no multi-agent routing needed for one clear task)

Tool: `search_docs(query)` -> ranked chunks from the product documentation

State: conversation history + last retrieval result, per session

Routing:

`call search_docs(user question)`

 if best chunk score < threshold:

 route to human handoff

 else:

 generate an answer grounded in the retrieved chunk(s)

Human handoff:

 package: the user's question + what was retrieved (if anything) + why it was flagged unsure

 send to: the support queue (exact destination depends on what the business already uses – ticketing, live-chat escalation, etc.)

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Advanced Version

Agent: single support agent

Tools:

`search_docs(query) -> ranked chunks + scores`

`escalate_to_human(conversation, reason) -> ticket id`

Grounding check (replaces naive self-confidence):

`retrieve chunks`

`if no chunk clears the similarity threshold: UNSURE -> escalate`

`else: generate the answer, but constrain the prompt to "answer only using the text below, say you don't know if it isn't there" – this catches the case where retrieval found something loosely related but not actually on-point`

Failure handling (Doc14 vocabulary):

`search_docs times out / errors -> don't crash, escalate with reason "docs search unavailable"`

`LLM call fails -> retry once, then escalate rather than retrying forever`

Monitoring:

`log every escalation with its reason (score-based miss vs. grounding-based miss) – this becomes a signal about gaps in the docs themselves, not just a bot-quality metric`

Scaling note (only if traffic actually requires it):

`cache repeated queries' retrieval results`

Difference between Basic, Intermediate, and Advanced: Basic is the one-paragraph plan anyone could sketch on a whiteboard in two minutes. Intermediate names the actual pieces — tool signature, state, routing condition, handoff payload — the way you'd actually write them in a design doc. Advanced adds what a naive design forgets entirely: what happens when the retrieval tool itself fails (not just when it finds nothing), a retry-then-escalate policy instead of an infinite retry loop, and turning escalations into a monitoring signal about the docs themselves, not just about the bot.

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